FORMATIVE ASSESSMENT – 2 (FA-2)

MATHEMATICS PROJECT WORK

Topic: Exponents and Powers

PRELIMINARY INFORMATION

Name of the Student: .....

Class: Class 8

Roll Number: .....

Name of the Lesson: Exponents and Powers

Name of the Guide Teacher: .....

AIM

To understand repeated multiplication, verify the fundamental laws of exponents with integral powers, and observe the systematic pattern of zero and negative exponents.

OBJECTIVES

1. To state and define the base, exponent, and essential laws of exponents.
2. To verify the Product Law and Quotient Law using simple integer bases.
3. To observe and represent the division pattern for zero and negative exponents.

MATERIAL REQUIRED

- A4 white paper / Project ruled sheet
- Geometry box tools (Ruler scale, pencil, eraser)
- Blue and black writing pens

PROCEDURE

Step 1: Understanding Exponents and Fundamental Laws

1. Exponential Form: In a^n, **a** is the base and **n** is the exponent or index (repeated multiplication).
Example: $2^4 = 2 \times 2 \times 2 \times 2 = 16$

2. Product Law: $a^m \times a^n = a^{m+n}$ (powers with the same base are added during multiplication).

3. Quotient Law: $a^m \div a^n = a^{m-n}$ (powers with the same base are subtracted during division).

4. Zero Exponent Law: Any non-zero number raised to exponent zero equals 1 ($a^0 = 1$).

5. Negative Exponent Law: $a^{-n} = {}^1/_{a^n}$ (multiplicative inverse of a^n).

PROCEDURE (CONTINUED)

Step 2: Verifying Laws of Exponents

Let us choose base **a = 2** with exponents **m = 3** and **n = 2**.

(i) Product Law Verification: [$2^3 \times 2^2 = 2^{3+2}$]

LHS: $2^3 \times 2^2 = (2 \times 2 \times 2) \times (2 \times 2) = 8 \times 4 = \mathbf{32}$

RHS: $2^{3+2} = 2^5 = 2 \times 2 \times 2 \times 2 \times 2 = \mathbf{32}$

✓ **Since LHS = RHS (32 = 32), the Product Law is verified.**

(ii) Quotient Law Verification: [$2^3 \div 2^2 = 2^{3-2}$]

LHS: $2^3 \div 2^2 = 8 \div 4 = \mathbf{2}$

RHS: $2^{3-2} = 2^1 = \mathbf{2}$

✓ **Since LHS = RHS (2 = 2), the Quotient Law is verified.**

(iii) Power of a Power Law Verification: [$(2^2)^3 = 2^{2 \times 3}$]

LHS: $(2^2)^3 = 4^3 = 4 \times 4 \times 4 = \mathbf{64}$

RHS: $2^{2 \times 3} = 2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = \mathbf{64}$

✓ **Since LHS = RHS (64 = 64), the Power of a Power Law is verified.**

(iv) Negative Exponent Verification: [$2^{-3} = {}^1/_{2^3}$]

Using Quotient Law: $2^0 \div 2^3 = 2^{0-3} = 2^{-3}$

In fractional form: ${}^{2^0}/_{2^3} = {}^1/_8 = {}^1/_{2^3}$

✓ **Hence, $2^{-3} = {}^1/_{2^3}$ is verified.**

CONCLUSION

- Exponents provide a compact and structured way to write repeated multiplication of the same base.
- The fundamental laws of exponents allow quick simplification of products and quotients.
- Negative exponents represent the reciprocal (multiplicative inverse) of positive powers, satisfying $a^{-n} = {}^1/_{a^n}$.

Signature of the Student: _____

Signature of the Guide Teacher: _____